

1 .Chemical And Corporate Identification

1.1 Product Marking

Product Name: Epoxy intumescent fire retardant coating CharCoat JF B Component

1.2.Recommended Uses and Prohibited Uses of Pure Substances or Mixtures

Intended Use: Jet-Fire Steel Structure Fire Protection For Professional Use Only.

construction Method: Please refer to the product technical data sheet.

1.3 Supplier information for the Safety Data Sheet.

Manufacturer: CharCoat Passive Fire Protection Inc. 220 Brew St Suite 301, Port Moody, BC V3H 0H6

Branch Contact Number: +16049411001 **email:** mail@chacoat.com

2. Hazard Overview

According to GB 13690-2009 and GB 30000-2013, substances or mixtures are classified as follows:

2.1 Classification of Pure Substances or Mixtures:

- Skin corrosion/irritation - Category 2
- Serious eye damage/eye irritation - Category 1
- Skin sensitization - Category 1
- Carcinogenicity - Category 2
- Acute aquatic toxicity - Category 3
- Chronic aquatic hazard - Category 3

2.2 Label Elements:

Using the toxicity data listed in Chapters 11 and 12, the product label is as follows.



2.3 Hazard Statements

Harmful if swallowed.

Causes serious eye irritation.

Causes skin irritation.

May cause an allergic skin reaction.

Toxic to aquatic life with long-lasting effects.

Precautions:

- Wear protective gloves.
- Wear protective goggles or a face shield.
- Avoid direct release into the environment.
- Avoid inhaling vapors.
- Do not eat, drink, or smoke while using this product.
- Thoroughly wash hands after handling.
- Contaminated work clothing should not be taken home.

Accident Response:

- Collect leaked material.
- If swallowed: If you feel unwell, call a poison control center or doctor, and rinse your mouth.
- In case of skin contact: Wash with plenty of soap and water, remove contaminated clothing and wash it before reuse.
- If skin irritation or a rash occurs: Seek medical attention.
- If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses if present and continue rinsing. If eye irritation persists: Get medical advice.

Safety Storage: N/A (Not Applicable).

Waste Disposal: Follow local regulations.

Additional information: Wear appropriate respiratory protection in case of inadequate ventilation.

2.4 Other Hazards

Not specified.

3. Composition Information:

This product is a mixture, primarily composed of:

Ingredient Name	Content	CAS
High Fatty Acid Polymer	≥ 25 -- ≤ 50	
Titanium Dioxide	≥ 10 -- ≤ 25	13463-67-7

Occupational Exposure Limits:

If there are occupational exposure limits, they will be listed in Section 8.

PBT substances or vPvB substances: Not included.

4. Emergency Measures

Description of necessary emergency measures.

4.1 General First Aid Measures

In case of any doubt or symptoms, seek medical attention. Do not feed anything through the mouth to an unconscious person.

Inhalation: Inhalation does not pose acute toxicity hazards. However, some solvents in the product may have mild toxicity. Inhaling vapors may cause irritation to the nose and throat, as well as neurological effects such as dizziness, nausea, headache, and drowsiness. If any condition arises, move the affected person to fresh air, keep warm, and avoid unnecessary movement. If breathing is irregular or stops, perform artificial respiration. Do not feed. Seek immediate medical attention.

Skin Contact: Prolonged skin contact may cause mild irritation. Remove contaminated clothing, wash thoroughly, and use soap and water to clean the affected area. If irritation persists, seek medical attention immediately.

Eye Contact: Direct eye contact may cause mild irritation. Immediately rinse eyes with plenty of water for at least 15 minutes, separating eyelids. Seek immediate medical attention. Chemical burns must be treated promptly by a doctor.

Ingestion: Seek immediate medical attention. Rinse mouth with water, remove dentures (if any), transfer the victim to fresh air, and maintain a calm breathing position. If the substance is swallowed, and the exposed person is conscious, drink a small amount of water. If the exposed person feels uncomfortable, stop and avoid causing a blockage of the airway with vomit, which can be dangerous. Unless instructed by medical personnel, do not induce vomiting. If vomiting occurs, keep the head low to prevent aspiration of vomitus into the lungs. Chemical burns must be treated promptly by a doctor. Do not feed anything through the mouth to an unconscious person. If unconscious, place the person in a recovery position, seek immediate medical attention, maintain an open airway, and loosen tight clothing, such as collars, ties, belts, or waistbands.

4.2 Most Important Symptoms and Effects, Including Acute and Chronic Effects

Potential Acute Health Effects:

Eye Contact: Causes severe eye irritation.

Inhalation: Exposure to decomposition products may be harmful to health.

Ingestion: Harmful if swallowed. Irritates the mouth, throat, and stomach.

Skin Contact: Causes skin irritation. May cause skin allergic reactions.

4.3 Symptoms of Overexposure

Eye Contact: Pain or irritation, tearing, redness.

Inhalation: No specific data.

Ingestion: No specific data.

Skin Contact: Adverse symptoms may include irritation, redness.

4.4 Emergency Medical First Aid Guidance and Necessary Special Treatment Measures

Physician's Notes:

If decomposition products are inhaled in a fire, symptoms may be delayed. Exposed individuals may need to be under medical observation for 48 hours.

Specific Treatment: No special treatment required.

First Aid Personnel Protective Measures:

Do not take first aid action if there is personal danger or if not properly trained. Performing mouth-to-mouth resuscitation on someone may be hazardous. Thoroughly wash with water or wear gloves before removing contaminated clothing.

5. Fire Fighting Measures

5.1 Fire Extinguishing Agents

Appropriate Extinguishing Agents:

Use extinguishing agents suitable for extinguishing surrounding fires.

Inappropriate Extinguishing Agents:

Not specified.

5.2 Special Hazards Arising from the Substance

Under combustion or heating conditions, there is a risk of increased pressure and container rupture.

The substance is toxic to aquatic life with long-lasting effects.

Collect and retain firefighting water contaminated by this material, and prevent it from being discharged into any waterway, sewer, or drain.

5.3 Hazardous Thermal Decomposition Products

Decomposition Products May Include The Following Substances:

Carbon dioxide
Carbon monoxide
Nitrogen oxides
Phosphorus oxides
Halogenated compounds

5.4 Advice for Firefighters

Special Protective Measures for Firefighters:

Evacuate all personnel from the accident site in case of a fire to quickly isolate the area. Do not take action if there is personal danger or if not properly trained.

Special Protective Equipment for Firefighters:

Firefighters should wear appropriate protective equipment and use self-contained breathing apparatus (SCBA) with full-face protection when facing a fire.

6. Leak Emergency Response

6.1 Personal Precautions, Protective Equipment, and Emergency Procedures

For Non-emergency Personnel:

Do not take action if there is any personal danger or if proper training has not been received.

Evacuate the surrounding area to prevent unnecessary exposure of unprotected individuals.

Do not touch or walk through spilled material; avoid breathing vapors or mist.

Provide adequate ventilation; wear appropriate respiratory protection in case of insufficient ventilation.

Wear suitable personal protective equipment.

For Emergency Responders:

If specific clothing is required to handle the leaked material, refer to Section 8 for relevant information on suitable and unsuitable materials.

Also, refer to "For Non-Emergency Personnel."

6.2 Environmental Precautions

Prevent the spread and runoff of spilled material, avoiding contact with soil, rivers, sewers, and drains.

Notify relevant authorities if the product causes environmental contamination (sewage, waterways, soil, or air).

Collect leaked material in case of a significant release.

6.3 Methods and Materials for Containment and Cleaning

Minor Spills:

If not hazardous, stop the leak and remove the container from the spill area.

Dilute with water, wipe if soluble in water, absorb if insoluble in water using inert absorbent material, and place it in an appropriate waste disposal container. Dispose through a licensed waste disposal contractor.

Major Spills:

If not hazardous, stop the leak and remove the container from the spill area.

Release upwind, preventing entry into sewers, drains, basements, or confined areas.

Flush the spilled material to a wastewater treatment plant or follow these steps:

Use non-combustible absorbent material (e.g., sand, soil, vermiculite, or diatomaceous earth) and place it in a container for disposal according to local regulations. Dispose through a licensed waste disposal contractor.

Contaminated absorbent material may pose hazards similar to the spilled product.

Note: Refer to Section 1 for emergency contact information and Section 13 for waste disposal.

7. Operation, Disposal, and Storage

7.1 Precautions for Safe Handling

Protective Measures:

Wear appropriate personal protective equipment (refer to Section 8).

Individuals with a history of skin allergies should not have direct contact with this product during any process.

Avoid contact with the eyes, skin, or clothing.

Do not swallow the product.

Avoid inhaling vapors or mist.

Prevent direct discharge into the environment.

Store in the original container or use approved alternative products made of compatible materials, and keep the container closed when not in use.

Residual product may remain in empty containers and may be hazardous.

Do not reuse discarded containers.

Dry sanding, flame cutting, and/or welding of dry paint films can produce dust and/or harmful fumes; wet sanding/smoothing should be used whenever possible. If exposure cannot be avoided through local exhaust ventilation, appropriate respiratory protective equipment should be used.

7.2 General Occupational Hygiene Advice

Smoking, eating, and drinking are prohibited in all preparation and application areas.

Workers should wash their hands and face before eating, drinking, or smoking.

Remove contaminated clothing and protective equipment before entering eating areas.

For additional hygiene measures, refer to Section 8.

7.3 Safety Storage Conditions, Including Incompatibles

Store according to local regulations.

Keep in a cool, dry area, away from frost, and avoid contact with incompatible materials (refer to Section 10) and food and beverages. Keep containers tightly closed and sealed before use; opened containers must be carefully resealed and stored upright to prevent leaks.

Do not store in unlabeled containers; use appropriate containers to avoid environmental contamination.

Keep away from the following materials: oxidizing agents, strong alkalis, and strong acids.

8. Exposure Controls and Personal Protection

8.1 Control Parameters

Occupational exposure limits are referenced from the Chinese National Occupational Health Standards (GBZ 2.1–2019 Occupational Exposure Limits for Hazardous Agents in the Workplace Part 1: Chemical Hazardous Agents).

Component	Exposure Limit (mg/cubic meter)	CAS Number	Remarks
Titanium Dioxide	8	13463-67-7	Dust and fumes

8.2 Control Measures

8.2.1 Engineering Controls

Adequate general ventilation should be sufficient to control workers' exposure to airborne contaminants.

8.2.2 Environmental Exposure Controls

Emissions from ventilation equipment or process equipment should be checked to ensure compliance with environmental protection regulations. In some cases, the use of scrubbers, filters, or engineering modifications to process equipment may be necessary to reduce emissions to acceptable levels.

8.2.3 Personal Protective Measures

Hygiene Measures

Thoroughly wash hands, forearms, and face after handling chemical products, before eating, smoking, using the toilet, and at the end of the work shift.

Contaminated clothing should be removed using appropriate techniques; contaminated work clothing should not be taken home, and before reuse, it should be cleaned.

Eye Protection

Safety glasses meeting approved standards designed to prevent liquid splashes should be used when the risk assessment indicates a need to avoid contact with liquid splashes, mists, gases, or dust.

Skin Protection

Hands: Use PVC or rubber gloves to minimize skin contact. Refer to the manufacturer's advice for selecting the most suitable gloves.

Body: Select personal protective equipment for the body based on the tasks being performed and the risks involved.

Other: Choose appropriate footwear and any other skin protection measures based on the tasks being performed and the associated risks to minimize skin contact.

Respiratory Protection

Use air-purifying or supplied-air respirators that comply with standards as needed, based on the tasks being performed and the risk assessment. The selection of respirators must be based on the known or anticipated levels of exposure, the hazards of the product, and the safe working limits of the chosen respirator.

9. Physical and Chemical Properties

Physical State: Liquid

Colour: White

Odor:	Odorless
Flash Point:	101°C (213.8°F)
Relative Density:	1.34
Solubility:	Insoluble in water
Viscosity:	Dynamic (at room temperature): 1309309 mm ² /s (1309309 cSt)

10. Stability and Reactivity

10.1 Chemical Stability

The product is stable under normal operating and storage conditions. Exposure to high temperatures may result in the formation of hazardous decomposition products such as carbon oxides, nitrogen oxides, and smoke.

10.2 Possible Hazardous Reactions

No hazardous reactions occur under normal storage and use conditions.

10.3 Materials to Avoid

Avoid contact with the following materials: oxidizing agents, strong bases, and strong acids.

10.4 Hazardous Decomposition Products

Under normal storage and use conditions, no hazardous decomposition products are produced.

11. Toxicological Information

11.1 Acute Toxicity

Repeated or prolonged exposure to this formulation may result in the loss of natural skin oils, leading to dryness, irritation, and potential non-allergic contact dermatitis. Liquid splashes into the eyes may cause reversible irritation and pain.

Acute toxicity testing has been conducted with the following results, indicating a toxic hazard:

Component	Route of Exposure	Test Subject	Dose
Titanium Dioxide	Inhalation of dust and smoke	Rabbit	6.82 mg/L for 4 hours

11.2 Irritation/Corrosion

Property	Category	Hazard
Skin Corrosion/Irritation	2	Causes skin irritation
Serious Eye Damage/Irritation	2	Causes serious eye irritation
Respiratory Sensitization	Unclassified	Not applicable
Skin Sensitization	Unclassified	Not applicable
Bacterial Toxicity	Unclassified	Not applicable
Carcinogenicity	Unclassified	Not applicable
Reproductive Toxicity	Unclassified	Not applicable
Teratogenicity	Unclassified	Not applicable
Specific Target Organ Toxicity (Single Exposure)	Unclassified	Not applicable
Specific Target Organ Toxicity (Repeated Exposure)	Not classified	Not applicable
Inhalation Hazard	Unclassified	Not applicable

11.3 Potential Acute Health Effects

Eye Contact: Causes serious eye irritation.

Inhalation: May release gases, vapors, or dust that are very irritating or corrosive to the respiratory system. Exposure to decomposition products may be harmful. Severe effects may be delayed after exposure.

Skin Contact: Causes skin irritation. May cause skin sensitization.

Ingestion: Harmful if swallowed, causing irritation of the mouth, throat, and stomach.

12. Ecological Information

12.1 Toxicity

The formulation was tested using the conventional method specified in Directive 1999/45/EC on dangerous substances. The results indicate that the substance is non-hazardous to the environment but contains substances harmful to the environment, and should not be discharged into sewers or rivers.

12.2 Persistence and Degradability

There is no available data on the persistence and degradability of the formulation itself.

12.3 Bioaccumulative Potential

No data is available regarding the potential for bioaccumulation.

12.4 Soil Mobility

There is no data on soil mobility.

12.5 PBT and vPvB Assessment Results

According to the assessment, the product does not contain any PBT/vPvB chemicals.

13. Waste Disposal

Efforts should be made to avoid or minimize the generation of waste. Disposal of this product and any by-products must always comply with environmental protection and waste disposal regulations, as well as the requirements of local authorities in any region.

Residual and non-recyclable products should be handled by licensed waste disposal contractors. Waste should not be discharged into sewers unless in full compliance with the requirements of all relevant regulatory authorities. Waste packaging should be recycled, with incineration or landfill considered only when recycling is not feasible.

Materials and containers must be handled in a safe manner, with caution taken when dealing with empty containers or liners that may retain some product residue. Spills should be contained to prevent spreading and contact with soil, rivers, sewers, and drainage channels.

14. Transport Information

14.1 Transport Hazard Class

Determine transportation conditions based on carrier requirements.

14.2 Packaging Category

Sealed; leakage is strictly prohibited.

14.3 Special Handling Instructions

Always transport in upright and securely sealed containers. For bulk transport, refer to the International Convention for the Control and Management of Ships' Ballast Water and Sediments (73/78 Annex II) and the International Maritime Organization's Code for the Construction and Equipment of Ships Carrying Dangerous Chemicals in Bulk.

15. Regulatory Information

This product complies with relevant domestic regulations.

16. Other Information

Version Information:

Issue Date/Revision Date: 2020-01-01

Reader Notice:

The information in this Chemical Safety Data Sheet is compiled based on our current level of understanding and current national laws. It does not guarantee the completeness and accuracy of this information. Users are responsible for verifying whether this data sheet is up-to-date before using related products. Users must make their own judgment on whether the product is suitable for their purposes. If these purposes are not specifically recommended in this data sheet, users assume the risk of using the product.